

Service Setup Guide (Windows)

[2 Tools Called]

Here's the quickest path to get the service running on your Windows machine and talking to all of its external dependencies.

1. Prereqs

- Install Python 3.12
 - Ensure Git is available.
 - Have credentials handy for Google Drive OAuth, MongoDB Atlas (or local MongoDB), Pinecone, and OpenAI.
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2. Clone & Enter the Repo

```
git clone <repo-url>
```

```
cd idp-etechtexas-rag
```

3. Create & Activate a Virtual Environment (PowerShell Example)

```
python -m venv .venv
```

```
.\.venv\Scripts\Activate.ps1
```

4. Install Dependencies

```
pip install --upgrade pip
```

```
pip install -r requirements.txt
```

5. Configure Environment Variables

- Copy `.env.example` if one exists, otherwise create `.env` in the repo root.
- Populate at least:
`ENV=local,`
`OPENAI_API_KEY,`
`GOOGLE_DRIVE_CLIENT_ID,`
`GOOGLE_DRIVE_CLIENT_SECRET,`
`GOOGLE_DRIVE_FOLDER_ID` (or override per request),
`MONGO_USERNAME,`
`MONGO_PASSWORD,`
`MONGO_CLUSTER_URL,`
`MONGODB_DATABASE,`
`PINECONE_API_KEY,`
`PINECONE_INDEX_NAME` (defaults to `idp-etechtexas-rag`),
`EMBEDDING_DIMENSION=1536,`
and any LangGraph tunables you want to change.
- Ensure `token.json` path in `.env` matches where you want the Google OAuth token cached (default already points to the root file here).

6. Provision External Services

- **Google Drive API:** create OAuth “Desktop App” credentials, download the client JSON, and feed values into `.env`. First run will pop a browser auth flow and persist `token.json`.
- **MongoDB:** have a cluster/database ready; allowlisting your IP or using `0.0.0.0/0` for testing.
- **Pinecone:** create the index (the app will auto-create if missing with dimension 1536).

- **OpenAI (or Azure OpenAI):** make sure the key is live; the project expects `text-embedding-3-small` and `GPT-4o-mini`.
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7. Launch the API

`uvicorn app.main:app --reload --host 0.0.0.0 --port 8000`

(Or `python -m app.main` if you prefer the built-in runner.)

8. Verify It's Up

- Visit <http://localhost:8000/health> → should report **OK**.
 - Swagger docs live at <http://localhost:8000/docs>.
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9. First OAuth Activation (Only Once Per Machine)

- Trigger any Google Drive endpoint (`/upload`, `/ocr/process`, `/ingestion/pipeline`).
 - Follow the browser prompt, approve scopes, and ensure `token.json` is saved.
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10. Optional UI

- Open `chatbot.html` directly in your browser or serve it from a static host.
 - Adjust any API base URLs inside if needed.
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11. Running the RAG Flows

- `/upload` → push files to Drive.

- `/ocr/process` or `/ingestion/pipeline` → OCR + metadata + embeddings.
 - `/chat` → LangGraph-driven QnA/summarization (ensure Pinecone + MongoDB contain your documents).
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12. Logs & Troubleshooting

- With `ENV=local`, check `logs/<date>/<time>.log`.
 - Common issues:
 - Missing credentials (400/401)
 - MongoDB network blocks
 - Pinecone dimension mismatch (set `EMBEDDING_DIMENSION=1536`)
 - Python 3.13 wheels (use 3.11/3.12)
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Next Steps

After you confirm the server boots:

- Seed a small test dataset through the pipeline.
- Inspect MongoDB/Pinecone for expected records.
- Drive a `/chat` request to ensure retrieval + generation work end-to-end.